Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Minimum answer = force per unit charge (accept force per coulomb)	1			1		
		(ii)	Use of field equation $E = \frac{kQ}{r^2}$ (1) Distance = $0.6\sqrt{2}$ **OR** $\frac{1.2}{\sqrt{2}}$ **OR** 0.85 [m] **OR** equivalent with sin **OR** cos45 (1) accept r^2 = 0.72 [m^2] Penultimate step seen **OR** 524 5XX seen for field (1) N.B. $k = 9 \times 10^9$ gives exactly 525 000	1	1 1		3	2	
		(iii)	Left (independent) **or** arrow pointing left (1) ÷ $\sqrt{2}$ seen **or** ×sin45 **or** ×cos45 (1) Final answer correct = 1 485 000 [N C^{-1}] (1)		3		3	2	
	(b)	(i)	Minimum answer – {work done / energy required} per unit charge from infinity (accept {work done / energy required} per coulomb from infinity)	1			1		
		(ii)	[Potential of] +ve and -ve charges cancel (1) Because equidistant **OR** because the same distance (1) accept P is at the centre **OR** mathematical: Potential equation used (1) An expression for 4 potentials seen to reduce to zero (k not needed) (1)	1	1		2		
		(iii)	Right and left force confirmed / agreed with (however stated e.g. accelerates right then decelerates is ok) (1) One of the two forces explained (we'll allow 2nd by implication) e.g. initially repulsed by -ve **OR** left force later because (closer to) +ve charges (1) E = 0 at infinity stated **or** explained somehow (1)			3	3		

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
		(iv)	Potential equation used (e.g. 630 000, -282 000) (1) Potentials added and subtracted (696 000) (1) WD = $q\Delta V$ used (± 0.0264) (1) Energy (not potential) equated to $\frac{1}{2}mv^2$ (1) Final answer (8.3 m s^{-1}) (1) **Alternative:** Potential equation used (-0.02394, +0.0107) (1) PEs added and subtracted (1) Value correct (± 0.0264) (1) Equated to $\frac{1}{2}mv^2$ (1) Final answer (8.3 m s^{-1}) (1) **Award a maximum of 3 marks for:** Any attempt at $a = \frac{F}{m}$ (1) expect a = 73.3) Any attempt at use of equations of motion (1) i.e. $v^2 = u^2 + 2ax$ Answer of 9.4 [m s^{-1}] (1) **Answers that score 4 marks:** 5.86 [m s^{-1}], 2 charges instead of 4 Power of 10 – cm instead of m should be 83 [m s^{-1}], same applies to mC instead of nC – out by a factor of 1000. Unfortunately, nC instead of μC will be out by a factor of $\sqrt{1000}$ = 31.6 [m s^{-1}] 7.46 [m s^{-1}] and 11.15[m s^{-1}]	1 1	 1 1 1		 5	 3	
			Question 4 total	**6**	**9**	**3**	**18**	**7**	**0**